

An AI-Powered System for Real-Time Fish Location Prediction for Small-Scale Fishermen

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Introduction: In recent years, small-scale and rural fishermen worldwide have struggled to locate fish efficiently due to unpredictable weather, changing sea currents, and limited access to modern technology. Traditional methods based on intuition are no longer reliable for securing consistent catches. This creates a pressing need for technological solutions that enhance fishing productivity and sustainability.

Objectives: The primary objective of this study is to design and implement an AI-powered system that predicts fish movements and suggests optimal fishing routes, specifically targeting fishermen in developing regions.

Methodology: The proposed system uses machine learning models, Random Forest and Long Short-Term Memory (LSTM) to analyze time-series environmental data. It integrates information from satellite imagery, weather APIs, tide and moon phase databases, and historical fishing records. The system also incorporates user feedback from fishermen to improve model accuracy over time. Data processing and predictions are delivered through a mobile application supporting both online and offline modes.

Results & Discussion: Field tests showed 85% to 95% accuracy in predicting fish-rich areas, leading to higher catch efficiency, reduced fuel consumption, and shorter fishing trips. Fishermen reported improved planning and fewer failed outings. The model successfully identified fish behavior patterns influenced by tide levels, moon phases, water temperature, noise, and feeding habits. The integration of user feedback further enhanced predictive performance. This approach not only modernizes traditional fishing but also supports ecological balance and rural development.

Conclusion: The AI-powered system offers a practical and scalable solution for improving fishing efficiency. While initial results are promising, further refinement and long-term impact assessments are recommended. The system holds potential to support rural economies, protect marine biodiversity, and foster technological advancement in traditional industries like fishing.

Keywords: Artificial Intelligence (AI), Machine Learning Models, Environmental Data Inputs, Real-Time Fish Location Prediction, Offline-Capable Mobile Application, Fisher Feedback Integration, Support for Rural and Small-Scale Fishermen